

## MYRNAM AGCM, AB, Canada

WMO: 710650

Lat: 53.7200N

Lon: 111.1144W

Elev: 1903

StdP: 13.71

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-----|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |     |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |     |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p) |
| 12            | -31.6      | -24.8 | -38.2                         | 0.7 | -31.5 | -31.0 | 1.0 | -24.6 | N/A                   | N/A  | N/A | N/A  | N/A                   | N/A  | N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 23.1                   | 82.5            | 63.5 | 79.6 | 62.0 | 76.7 | 61.0 | 66.7                | 78.1 | 64.8 | 75.6 | 63.0 | 73.1 | N/A                  | N/A  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 62.5                            | 90.9 | 72.7 | 60.5 | 84.4 | 70.2 | 58.7 | 79.1 | 68.1 | 32.4          | 78.3 | 30.9 | 75.5 | 29.4 | 73.0 | 74.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) N/A           | N/A  | N/A | -40.7                      | 88.2 | 3.9                | 3.3 | -43.5                                              | 90.6 | -45.8      | 92.5 | -48.0      | 94.4 | -50.9      | 96.8 |    |    |
| (5)               |      |     | -40.9                      | 71.3 | 3.9                | 2.3 | -43.7                                              | 73.0 | -46.0      | 74.4 | -48.2      | 75.7 | -51.1      | 77.4 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 35.2   | 8.9   | 7.7   | 21.3  | 36.6  | 50.2  | 58.2  | 61.9  | 59.6  | 50.5  | 37.8  | 20.4  | 7.5   |  |  |
|                                                                        | DBStd       | 22.91  | 15.61 | 14.77 | 13.59 | 9.51  | 7.85  | 5.60  | 4.64  | 5.65  | 7.57  | 8.24  | 12.30 | 14.84 |  |  |
|                                                                        | HDD50       | 6543   | 1273  | 1185  | 891   | 411   | 97    | 8     | 0     | 4     | 83    | 386   | 888   | 1317  |  |  |
|                                                                        | HDD65       | 10921  | 1738  | 1605  | 1356  | 853   | 461   | 212   | 118   | 181   | 435   | 842   | 1338  | 1782  |  |  |
|                                                                        | CDD50       | 1144   | 0     | 0     | 0     | 8     | 102   | 254   | 370   | 302   | 99    | 9     | 0     | 0     |  |  |
|                                                                        | CDD65       | 46     | 0     | 0     | 0     | 0     | 1     | 7     | 22    | 15    | 1     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 1251   | 0     | 0     | 0     | 6     | 128   | 204   | 418   | 367   | 123   | 5     | 0     | 0     |  |  |
|                                                                        | CDH80       | 221    | 0     | 0     | 0     | 1     | 21    | 32    | 80    | 67    | 20    | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 16.3   | 0.7   | 0.5   | 0.6   | 1.2   | 1.7   | 2.8   | 3.0   | 2.3   | 1.3   | 0.7   | 0.7   | 0.7   |  |  |
|                                                                        | PrecMax     | 22.1   | 2.4   | 0.9   | 1.5   | 4.1   | 3.7   | 5.5   | 6.5   | 4.6   | 3.1   | 1.8   | 3.0   | 1.3   |  |  |
|                                                                        | PrecMin     | 8.2    | 0.2   | 0.1   | 0.2   | 0.1   | 0.2   | 0.6   | 1.0   | 0.5   | 0.0   | 0.1   | 0.1   | 0.2   |  |  |
|                                                                        | PrecStd     | 3.1    | 0.5   | 0.2   | 0.3   | 0.8   | 0.9   | 1.1   | 1.2   | 0.9   | 0.8   | 0.4   | 0.6   | 0.3   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 45.1  | 42.8  | 54.4  | 72.8  | 82.9  | 83.8  | 86.2  | 85.9  | 82.8  | 71.7  | 51.4  | 42.4  |  |  |
|                                                                        |             | MCWB   | 38.0  | 36.8  | 43.1  | 50.5  | 56.7  | 62.7  | 67.2  | 65.3  | 60.9  | 54.1  | 41.3  | 35.3  |  |  |
|                                                                        | 2%          | DB     | 40.5  | 38.6  | 47.6  | 64.9  | 77.8  | 79.6  | 82.0  | 81.4  | 77.7  | 62.5  | 44.7  | 37.6  |  |  |
|                                                                        |             | MCWB   | 35.1  | 33.9  | 39.1  | 46.9  | 55.1  | 60.8  | 66.1  | 64.4  | 58.5  | 49.7  | 38.0  | 32.2  |  |  |
|                                                                        | 5%          | DB     | 37.0  | 34.4  | 42.6  | 58.7  | 72.9  | 75.7  | 78.5  | 78.1  | 73.4  | 57.2  | 39.7  | 33.0  |  |  |
|                                                                        |             | MCWB   | 33.0  | 31.3  | 36.4  | 44.3  | 53.3  | 59.0  | 64.2  | 63.2  | 57.0  | 46.2  | 34.9  | 29.5  |  |  |
|                                                                        | 10%         | DB     | 32.6  | 30.9  | 38.7  | 52.9  | 68.6  | 72.3  | 75.4  | 74.9  | 67.8  | 52.5  | 36.3  | 28.6  |  |  |
|                                                                        |             | MCWB   | 29.5  | 28.2  | 33.8  | 41.6  | 51.1  | 57.1  | 62.7  | 61.6  | 54.6  | 44.0  | 32.9  | 26.2  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 38.3  | 37.6  | 44.1  | 51.2  | 59.0  | 65.6  | 70.8  | 69.1  | 62.4  | 55.0  | 41.8  | 36.7  |  |  |
|                                                                        |             | MCDB   | 44.1  | 41.9  | 53.2  | 71.1  | 75.8  | 79.7  | 82.8  | 80.9  | 75.6  | 69.1  | 50.0  | 42.5  |  |  |
|                                                                        | 2%          | WB     | 35.7  | 34.2  | 39.6  | 47.5  | 56.5  | 62.9  | 67.6  | 66.3  | 59.7  | 50.9  | 38.3  | 32.4  |  |  |
|                                                                        |             | MCDB   | 40.2  | 37.3  | 46.9  | 62.9  | 72.1  | 74.4  | 78.9  | 78.3  | 74.5  | 59.5  | 43.3  | 36.9  |  |  |
|                                                                        | 5%          | WB     | 33.1  | 31.6  | 36.7  | 44.9  | 54.7  | 61.0  | 65.6  | 64.3  | 57.6  | 47.8  | 35.1  | 29.6  |  |  |
|                                                                        |             | MCDB   | 37.0  | 34.7  | 41.9  | 57.4  | 69.6  | 71.9  | 76.1  | 74.8  | 71.4  | 55.8  | 38.6  | 32.8  |  |  |
|                                                                        | 10%         | WB     | 29.4  | 27.9  | 34.3  | 42.2  | 52.7  | 59.3  | 63.8  | 62.5  | 55.4  | 44.7  | 33.1  | 26.2  |  |  |
|                                                                        |             | MCDB   | 32.5  | 30.9  | 38.4  | 51.5  | 66.5  | 69.4  | 73.4  | 73.1  | 66.2  | 52.7  | 36.2  | 28.6  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 21.2  | 22.8  | 20.5  | 21.4  | 27.0  | 23.2  | 23.1  | 25.5  | 27.0  | 21.5  | 17.8  | 18.8  |  |  |
|                                                                        |             | MCDBR  | 22.9  | 25.5  | 24.3  | 32.7  | 35.0  | 30.3  | 29.8  | 31.7  | 37.8  | 32.0  | 22.1  | 23.5  |  |  |
|                                                                        |             | MCWBR  | 18.4  | 21.5  | 16.8  | 17.1  | 15.9  | 14.0  | 15.8  | 16.5  | 19.7  | 19.3  | 16.3  | 19.0  |  |  |
|                                                                        | 5% WB       | MCDBR  | 22.6  | 23.7  | 21.7  | 30.7  | 30.9  | 25.3  | 26.5  | 28.4  | 34.9  | 29.4  | 19.8  | 22.8  |  |  |
|                                                                        |             | MCWBR  | 18.6  | 20.2  | 15.2  | 17.0  | 14.4  | 12.8  | 14.7  | 15.8  | 18.9  | 18.5  | 15.1  | 18.9  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.249 | 0.285 | 0.294 | 0.333 | 0.352 | 0.370 | 0.367 | 0.378 | 0.313 | 0.277 | 0.251 | 0.219 |  |  |
|                                                                        | taud        |        | 2.305 | 2.234 | 2.345 | 2.359 | 2.356 | 2.331 | 2.371 | 2.311 | 2.484 | 2.556 | 2.445 | 2.352 |  |  |
|                                                                        | Ebn at Noon |        | 235   | 258   | 284   | 285   | 284   | 279   | 277   | 266   | 275   | 266   | 238   | 230   |  |  |
|                                                                        | Edh at Noon |        | 18    | 27    | 30    | 34    | 36    | 38    | 36    | 36    | 26    | 19    | 16    | 14    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 301   | 635   | 1115  | 1484  | 1839  | 1877  | 1931  | 1575  | 1108  | 633   | 301   | 221   |  |  |
|                                                                        | RadStd      |        | 32    | 47    | 86    | 128   | 134   | 135   | 105   | 118   | 121   | 92    | 34    | 22    |  |  |

## Historical Trends

|                              |     | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------------------------------|-----|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|                              |     |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
| (a)                          | (b) | (c)   | (d)     | (e)    | (f)   | (g)     | (h)   | (i) | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |     |       | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (63 neighbors) |     |       | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page